

### 3.1.5 Kinetics

The study of kinetics enables chemists to determine how a change in conditions affects the speed of a chemical reaction. Whilst the reactivity of chemicals is a significant factor in how fast chemical reactions proceed, there are variables that can be manipulated in order to speed them up or slow them down.

### 3.1.5.1 Collision theory

| Content                                                                                                                                                                                                                                                                                                                                               | Opportunities for skills development |
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| <p>Reactions can only occur when collisions take place between particles having sufficient energy.</p> <p>This energy is called the activation energy.</p> <p><b>Students should be able to:</b></p> <ul style="list-style-type: none"><li>define the term activation energy</li><li>explain why most collisions do not lead to a reaction.</li></ul> |                                      |

### 3.1.5.2 Maxwell–Boltzmann distribution

| Content                                                                                                                                                                                                                            | Opportunities for skills development |
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| <p>Maxwell–Boltzmann distribution of molecular energies in gases.</p> <p><b>Students should be able to:</b></p> <ul style="list-style-type: none"><li>draw and interpret distribution curves for different temperatures.</li></ul> |                                      |

### 3.1.5.3 Effect of temperature on reaction rate

| Content                                                                                                                                                                                                                                                                                                                                       | Opportunities for skills development                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <p>Meaning of the term rate of reaction.</p> <p>The qualitative effect of temperature changes on the rate of reaction.</p> <p><b>Students should be able to:</b></p> <ul style="list-style-type: none"><li>use the Maxwell–Boltzmann distribution to explain why a small temperature increase can lead to a large increase in rate.</li></ul> | <p><b>AT a, b, k and l</b></p> <p><b>PS 2.4 and 3.1</b></p> <p>Students could investigate the effect of temperature on the rate of reaction of sodium thiosulfate and hydrochloric acid by an initial rate method.</p> <p><b>Research opportunity</b></p> <p>Students could investigate how knowledge and understanding of the factors that affect the rate of chemical reaction have changed methods of storage and cooking of food.</p> |
| <p><b>Required practical 3</b></p> <p>Investigation of how the rate of a reaction changes with temperature.</p>                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### 3.1.5.4 Effect of concentration and pressure

| Content                                                                                                                                                                                                                                                                                                                                                                   | Opportunities for skills development                                                                                                                                                                                      |
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| <p>The qualitative effect of changes in concentration on collision frequency.</p> <p>The qualitative effect of a change in the pressure of a gas on collision frequency.</p> <p><b>Students should be able to:</b></p> <ul style="list-style-type: none"><li>• explain how a change in concentration or a change in pressure influences the rate of a reaction.</li></ul> | <p><b>AT a, e, k and i</b></p> <p>Students could investigate the effect of changing the concentration of acid on the rate of a reaction of calcium carbonate and hydrochloric acid by a continuous monitoring method.</p> |

### 3.1.5.5 Catalysts

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Opportunities for skills development |
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| <p>A catalyst is a substance that increases the rate of a chemical reaction without being changed in chemical composition or amount.</p> <p>Catalysts work by providing an alternative reaction route of lower activation energy.</p> <p><b>Students should be able to:</b></p> <ul style="list-style-type: none"><li>• use a Maxwell–Boltzmann distribution to help explain how a catalyst increases the rate of a reaction involving a gas.</li></ul> |                                      |